

PES UNIVERSITY · SOFTWARE ENGINEERING LAB

Lab 3: Component Modelling & Architectural Pattern Selection

Evaluation of Architectural Patterns, Component Decomposition & Interface Modelling

Student Name:	Arjun Srikanth	SRN: PES1UG24AM920
Class / Section:	5th Sem B.Tech CSE · Section H	Lab Duration: 90 Minutes
Assigned Problem Statement:	#48 · Incident Escalation & On-Call Rotation Engine	
Handout Scenario:	Self-Service Coffee Kiosk System	
GitHub Repository:	github.com/arjunsrikanthh/PES1UG24AM920_se_lab	

1. Comparative Analysis of Architectural Styles

Pattern	Structural Organization	Key Advantages (Pros)	Key Limitations (Cons)
Layered (N-Tier)	Organized in horizontal layers: Presentation, Business Logic, and Data Access with unidirectional dependency.	? Clear separation of concerns ? Intuitive and easy to develop ? Straightforward localized testing ? Low infrastructure complexity	? Performance overhead across layers ? Monolithic blast radius ? Inflexible uniform scaling ? Hard to scale individual components
Microservices Architecture	Fine-grained, independently deployable services centered around business domains; interact via gRPC/REST/broker.	? Fault isolation prevents cascades ? Targeted horizontal autoscaling ? Independent deployments ? Polyglot technology diversity	? High operational/network overhead ? Eventual consistency complexity ? Distributed debugging & telemetry ? Network partition overhead
Client-Server Architecture	Centralized server node handling business logic & state, serving distributed light/rich client nodes.	? Centralized rule governance ? Guaranteed ACID consistency ? Simpler client endpoint compute ? Straightforward administration	? Central server is a Single Point of Failure ? Scalability bottleneck under surge ? Network failure locks client access ? Tight client-server coupling

Analytical Synthesis: For distributed high-availability incident management (Problem Statement #48), Microservices Architecture is selected to guarantee fault isolation and independent scaling during alert storms. For the standalone Self-Service Coffee Kiosk, Layered Component Architecture is selected for direct peripheral access, hardware sandboxing, and zero-latency local execution.

Incident Escalation & On-Call Rotation Engine · Architecture Selection & Component Model

Architecture Selection Statement: "We chose an Event-Driven Microservices Architecture for the Incident Escalation Engine."



Component Name	Stereotype	Provided Interfaces (Ball)	Required Interfaces (Socket)
Webhook Ingestion Gateway	<<component>>	AlertWebhook (HTTPS POST)	IEscalationControl (Event Bus)
Rotation & Schedule Manager	<<component>>	IScheduleLookup, IRotationConfig	ISqlPersistence (Postgres TLS)
Escalation & Rules Engine	<<component>>	IEscalationControl (RPC/Timer)	IScheduleLookup, INotifyDispatch
Multi-Channel Notification Svc	<<component>>	INotifyDispatch (Async Queue)	ITelecomGateway (Twilio API)
Triage & Lifecycle Manager	<<component>>	IDashboardAPI, IRemoteTriageWebhook	IEscalationControl, IPostMortemTrigger
Post-Mortem & Telemetry Store	<<component>>	IPostMortemTrigger (Event)	ISqlPersistence (Audit Store)

3. Deliverable B: Handout Scenario (Self-Service Coffee Kiosk System)

Self-Service Coffee Kiosk System · Architecture Selection & Component Model

Architecture Selection Statement: "We chose a Layered Component-Based Architecture for the Coffee Kiosk System."

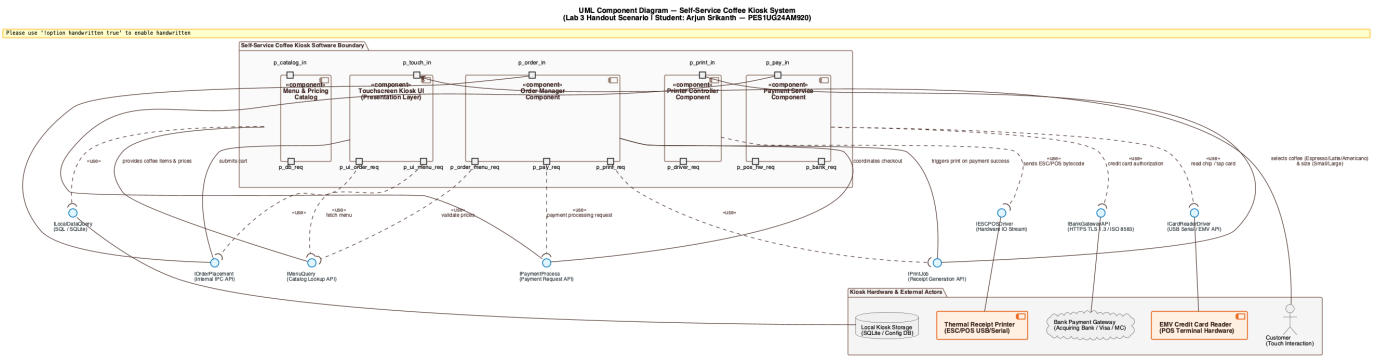


Figure 2: UML Component Diagram · Self-Service Coffee Kiosk System (5 Components, 8 Ball/Socket Interfaces)

Component Name	Stereotype	Provided Interfaces (Ball)	Required Interfaces (Socket)
Touchscreen Kiosk UI	<<component>>	.	IMenuQuery, IOrderPlacement
Order Manager (Given)	<<component>>	IOrderPlacement	IMenuQuery, IPaymentProcess, IPrintJob
Payment Service (Given)	<<component>>	IPaymentProcess	ICardReaderDriver, IBankGatewayAPI
Menu & Pricing Catalog	<<component>>	IMenuQuery	ILocalDataQuery (SQLite)
Printer Controller	<<component>>	IPrintJob	IESCPOSDriver (USB Serial)

4. UML Component Notation & Deliverable Compliance Summary

Verification against Lab 3 Handout Specifications and Academic Standards

4.1 UML 2.5 Component Modelling Standards Applied:

1. Component Rectangles & Stereotypes: All components are encapsulated black-box entities designated with the <<component>> stereotype and standard upper-right icon symbol. Internal implementations are strictly hidden behind interfaces.
2. Provided Interfaces (Ball / Lollipop Notation): Represent services offered by a component to other modules. Illustrated as a solid line originating from the component ending in a complete circle (e.g., IOrderPlacement, IScheduleLookup).
3. Required Interfaces (Socket Notation): Represent external services and dependencies required by a component to execute its functions. Illustrated as a semicircle enclosing or connecting to the provided interface ball (e.g., IBankGatewayAPI, INotifyDispatch).
4. Explicit Ports: Ports are drawn along the component boundaries (e.g., p_webhook_in, p_pay_in) to group cohesive interfaces and establish explicit interaction boundaries between components and external hardware/network layers.
5. Usage Dependencies & Assembly Connectors: Dashed directed arrows labeled <<use>> signify compile-time or runtime behavioral dependencies, while assembly connectors join provided and required interfaces directly without intermediary wiring.

4.2 Deliverables Compliance Checklist

Requirement / Deliverable	Handout Specification	Achieved Status	Compliance Verification
Component Count	>= 5 components	6 (PS #48) / 5 (Coffee)	Full compliance with <<component>> syntax
Interface Count	>= 4 interfaces	8 (PS #48) / 8 (Coffee)	Full compliance with Ball & Socket notation
Given Components	Order Manager & Payment Svc	Included & Modeled	Both explicitly present in Coffee Kiosk model
Architectural Styles	Compare Layered, Microservices, C/S	Completed (Table 1)	Detailed trade-off matrix in Section 1
Selection Justification	Choice, 2 Reasons, Sec, Perf	Completed (Sec 2 & 3)	Scenario-specific rationales documented
Export Formats	PNG and PDF diagrams + Word doc	All Generated	PUML, PNG, Vector PDF, DOCX, & PDF report
Git Repository Sync	Push to student GitHub repo	Synchronized	Pushed to arjunsrikanthh/PES1UG24AM920_se_lab

Academic Integrity & Submission Statement:

I hereby certify that the architectural analysis, component models, interface definitions, and justification documents submitted in this report are original work completed for SE Lab 3 (PES University, Bengaluru). All diagrams follow ISO/IEC UML 2.5 specifications.